Maynooth University
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EE202 : Introduction to Telecommunications

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Section 4: Formatting and Baseband Modulation

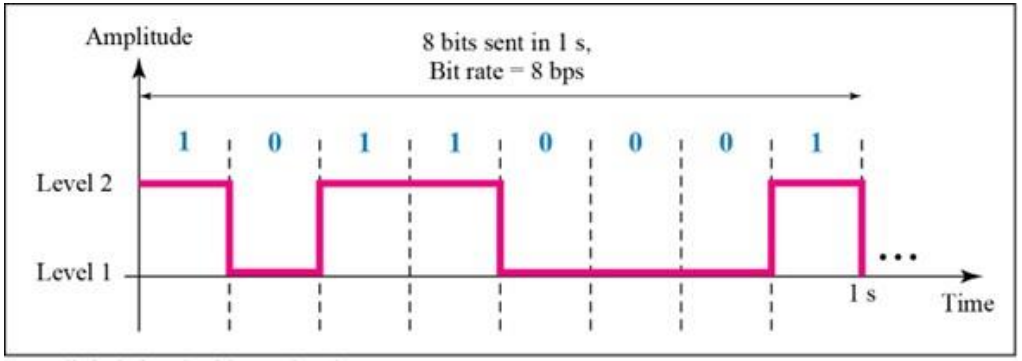
Data Transmission

Baseband Transmission

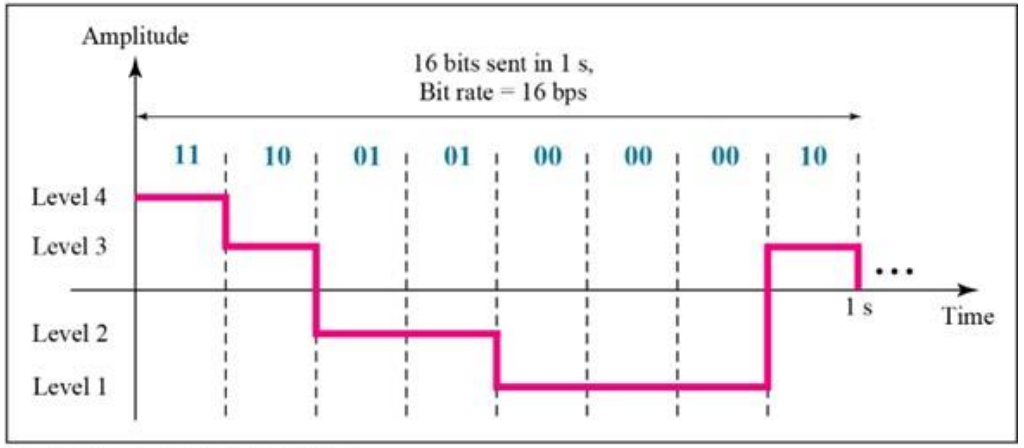
Baseband Digital Transmission

Recap on Digital Signals

Two digital signals: one with two signal levels and the other with four signal levels



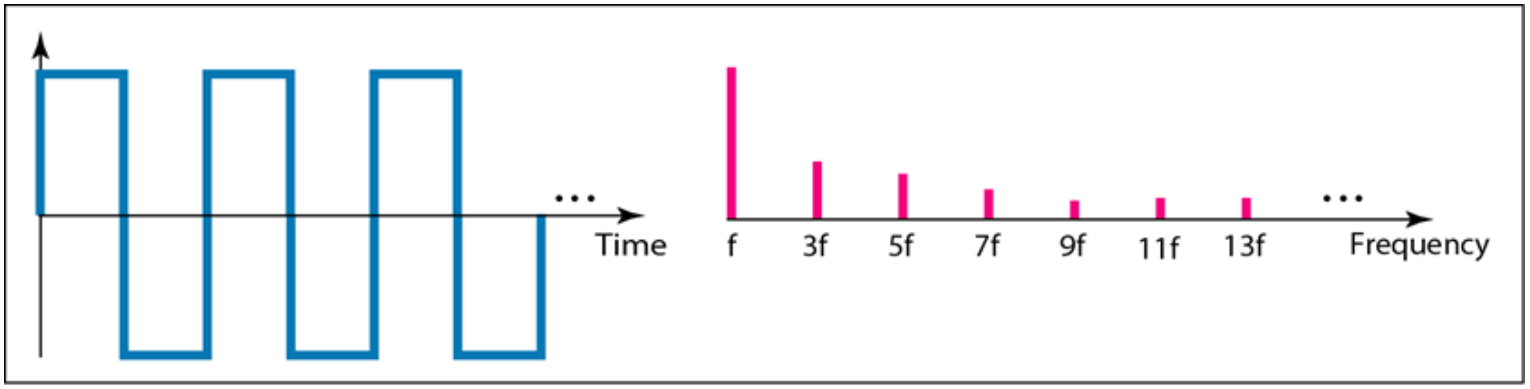
a. A digital signal with two levels



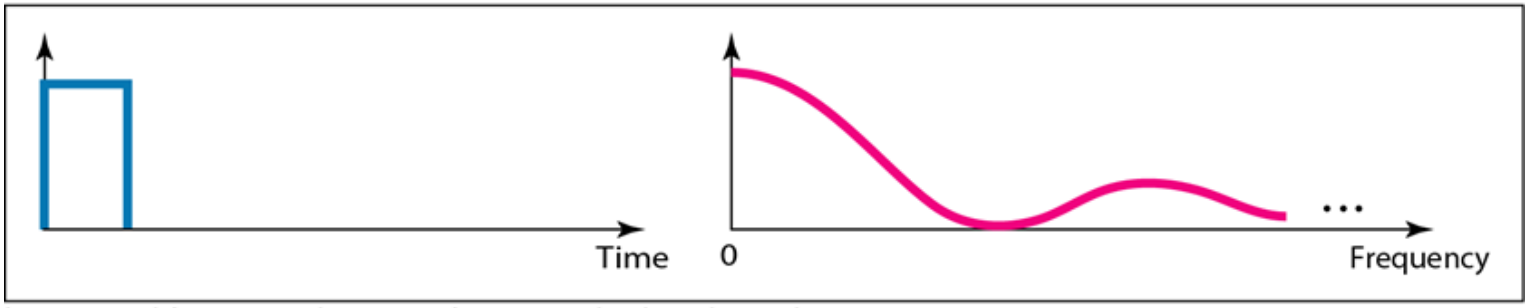
b. A digital signal with four levels

Recap on Digital Signals

The time and frequency domains of periodic and aperiodic digital signals.



a. Time and frequency domains of periodic digital signal



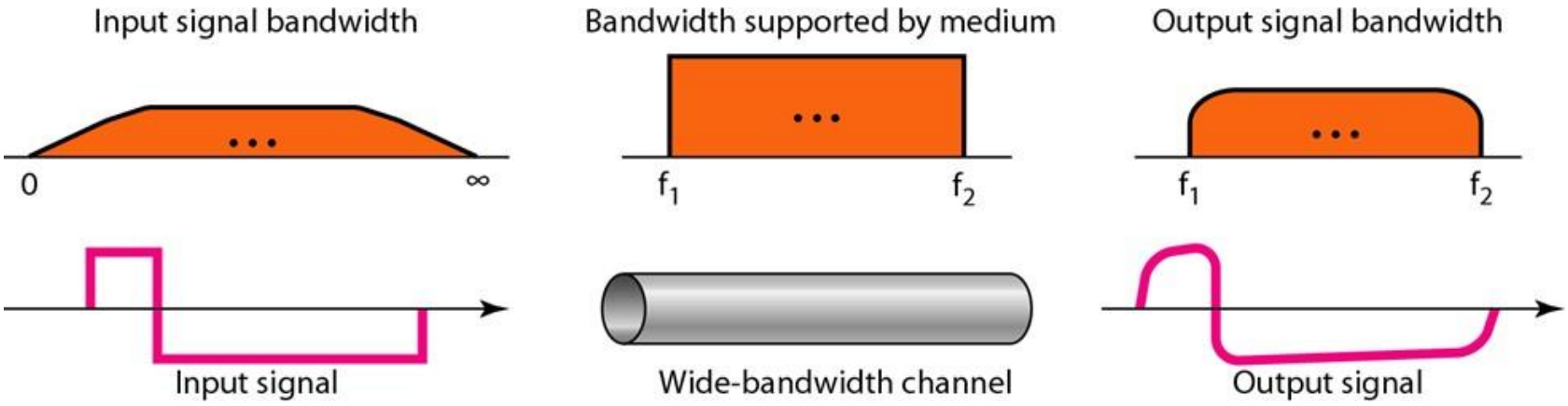
b. Time and frequency domains of nonperiodic digital signal

Signal Transmission

- For signal transmission the first step is to do formatting to ensure that the message (or source signal) is compatible with digital processing.
- **Transmit formatting** is a transformation from source information to digital symbols.
- When data compression in addition to formatting is employed, the process is termed as **source coding**.
- Signal transmission can be achieved through Baseband transmission and Passband transmission.

Baseband Transmission

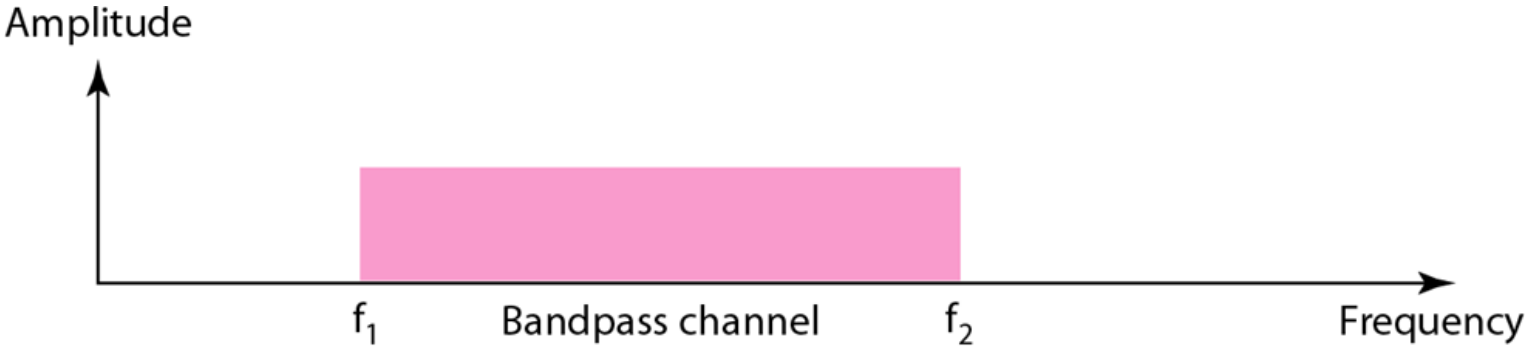
A digital signal is a composite analog signal with an infinite bandwidth.



Baseband transmission of a digital signal that preserves the shape of the digital signal is possible only if we have a low-pass channel with an infinite or very wide bandwidth.

Passband Transmission

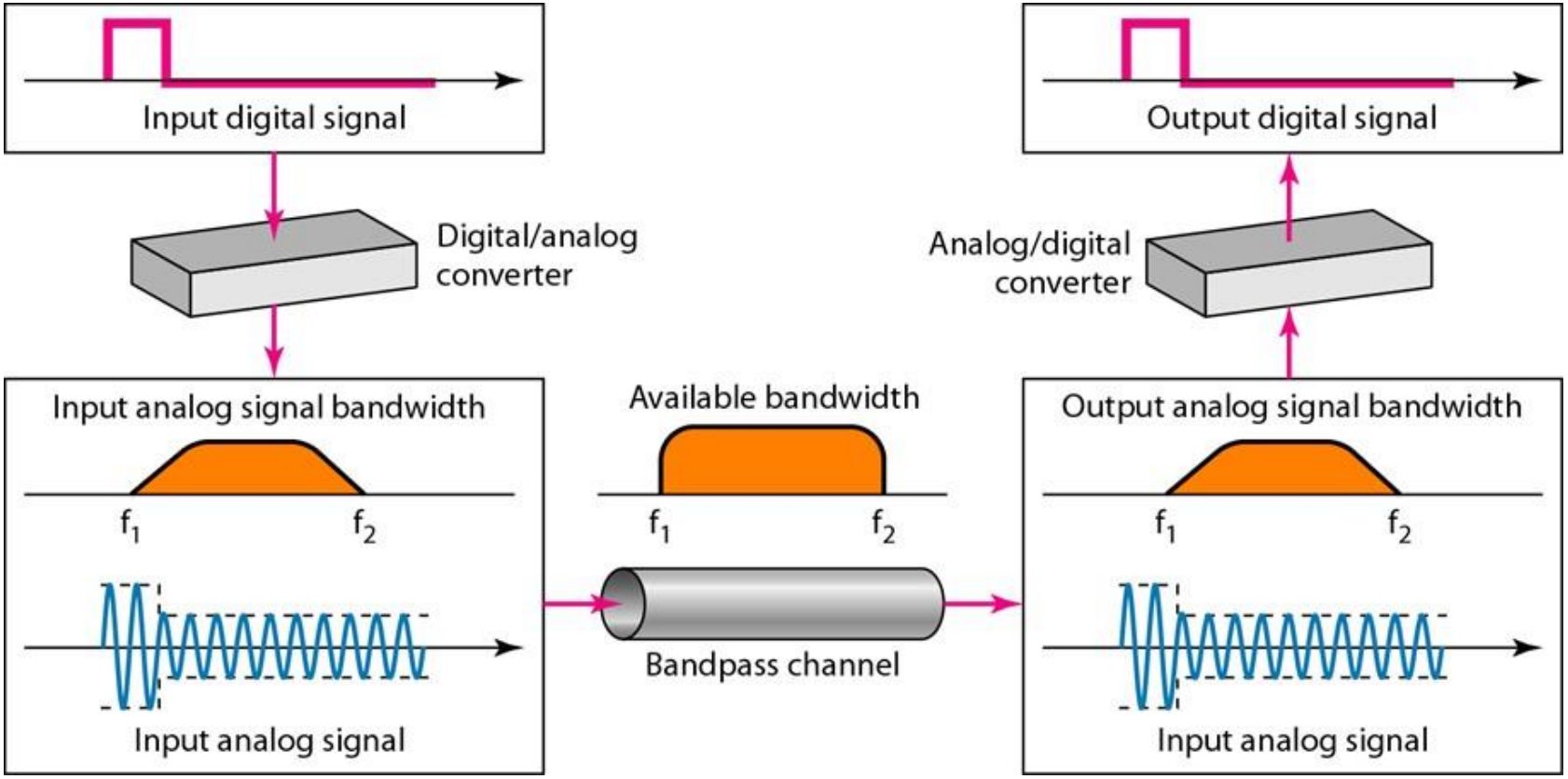
In baseband transmission, the required bandwidth is proportional to the bit rate; if we need to send bits faster, we need more bandwidth.



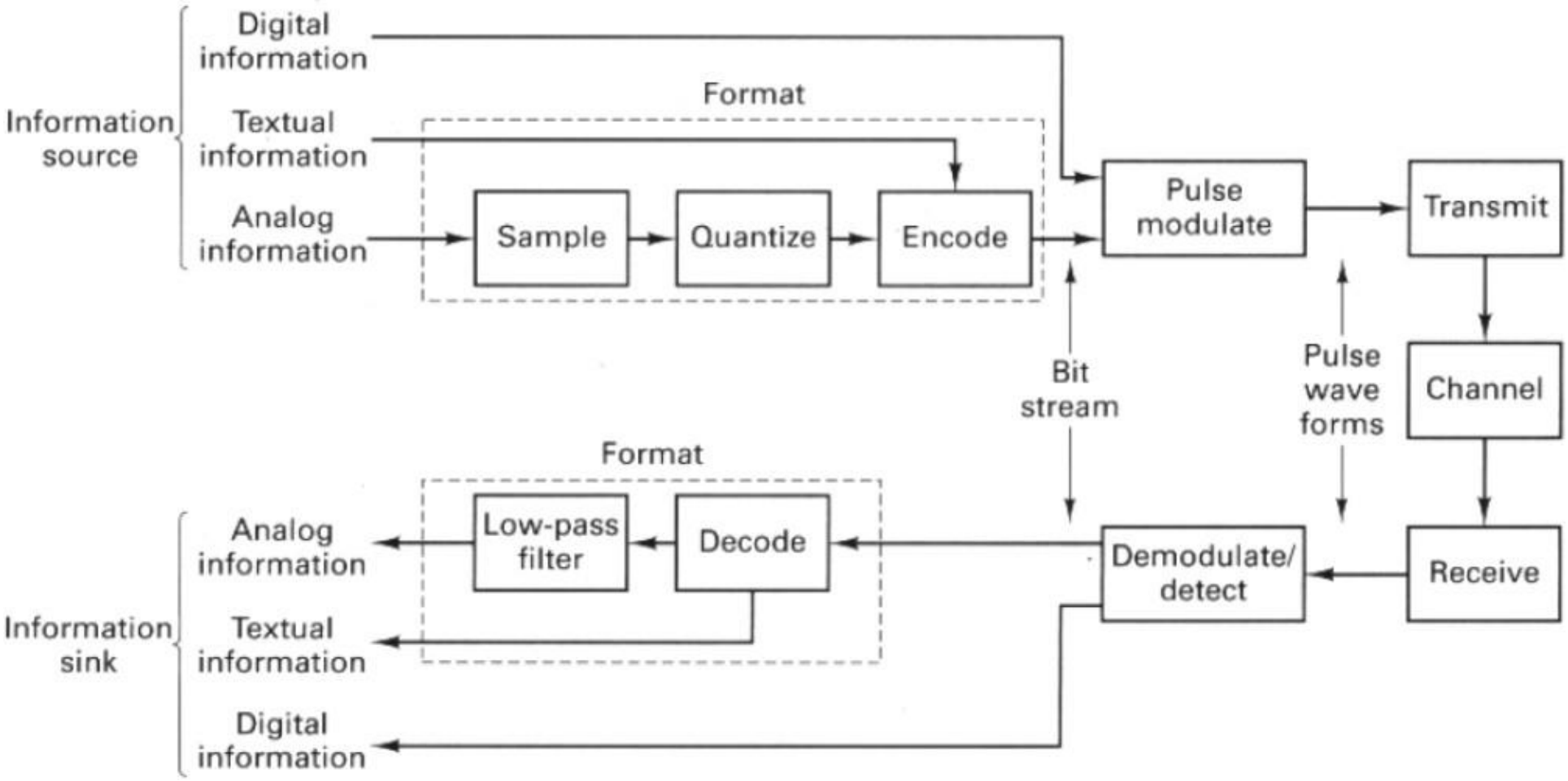
If the available channel is a bandpass channel, we cannot send the digital signal directly to the channel; we need to convert the digital signal to an analog signal before transmission.

Passband Transmission

Modulation of a digital signal for transmission on a bandpass channel



Formatting for Baseband Transmission



A digital signal is superior to an analog signal because it is more robust to noise and can easily be recovered, corrected and amplified. For this reason, the tendency today is to change an analog signal to digital data.
In this section we describe two techniques,

- Pulse Code Modulation (PCM)
- Delta Modulation (DM)

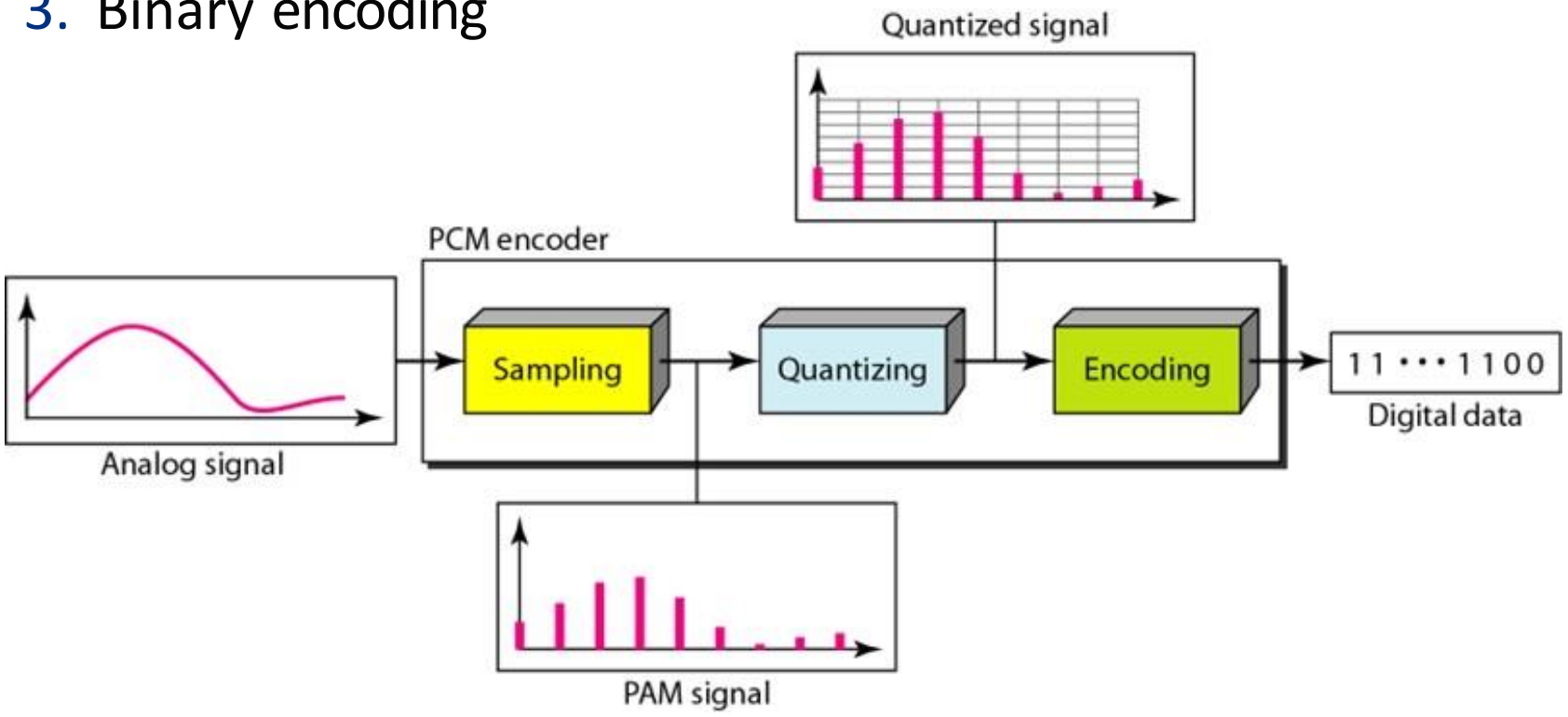
Encoding Digital Signals: PCM

- This is the process where we take an analogue signal, quantize it and represent it in digital form, whether binary or more complex schemes.
- It is NOT a modulation scheme in the sense of PWM, PDM, PPM, which are described at then end of this lecture. PCM just says the data is in digital form. When it is used as a term, the data is encoded to digital form as :
  - 8 - 24 bit resolution
  - 8 - 192 kHz sampling rate
- It has NO pulses. There is no reason for the name other than it was developed at a time when PWM/PPM were very common and pulse was what every digital modulation scheme needed.

Pulse Coded Modulation

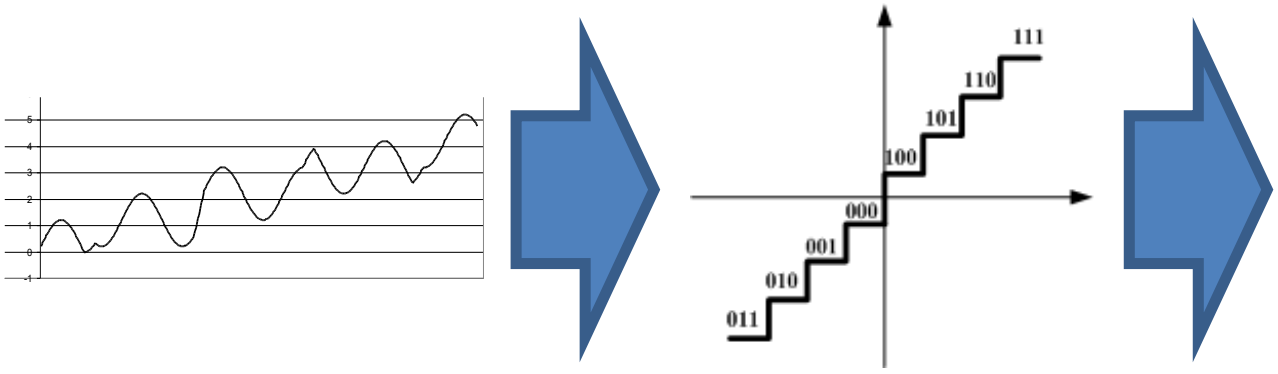
PCM consists of three steps to digitize an analog signal:

1. Sampling
2. Quantization
3. Binary encoding



PCM Example

- We decide to encode the analogue signal using $n = 3$ bits.
- This means that we can quantize the analogue signal with $L = 2^3 = 8$ different values.
- Each one of these values will be given a 3-bit binary code (000, 001, . . . , 111).
- The higher the number of bits, the larger the number of finite levels, and therefore the resolution.

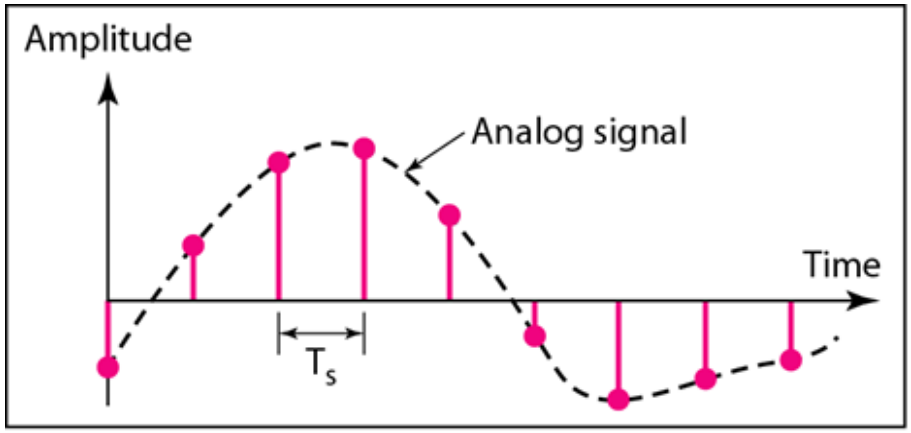


010, 110, 011, 001, 110, …..

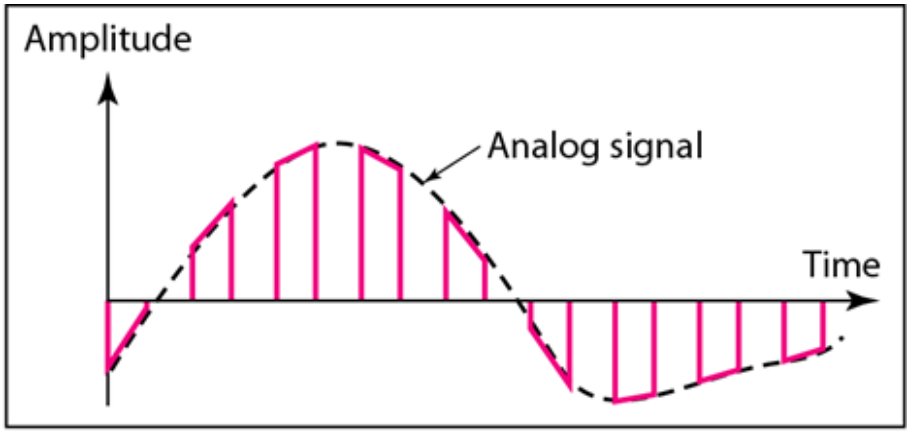
Sampling

- Analog signal is sampled every T_s sec, T_s is referred to as the sampling interval. $f_s = 1/T_s$ is called the sampling rate or sampling frequency.
- There are 3 sampling methods:
  - *Ideal* - an impulse at each sampling instant
  - *Natural* - a pulse of short width with varying amplitude
  - *Flat-top* - sample and hold, like natural but with single amplitude value
- The output of the Natural sampling process is often referred to as Pulse Amplitude Modulation (PAM) because the successive output intervals can be described as a sequence of pulses with amplitudes derived from the input waveform samples.

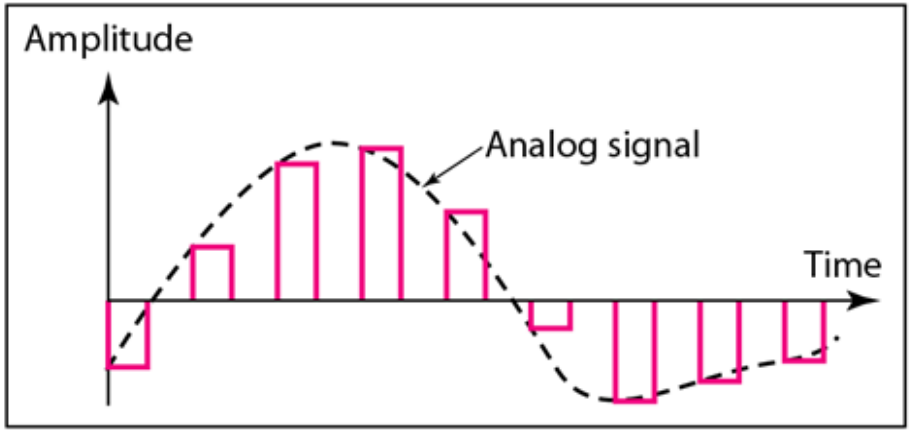
Sampling Methods for PCM



a. Ideal sampling



b. Natural sampling



c. Flat-top sampling

Nyquist Criterion

- A band-limited signal having no spectral components above f_m Hz can be determined uniquely by values sampled at uniform intervals of $T_s \leq \frac{1}{2f_m}$ sec.
- If $f_s = 1/T_s$, the sampling rate should satisfy,
  - $f_s \geq 2f_m$ - *Nyquist Criterion*
  - $f_s = 2f_m$ - *Nyquist Rate*
- Nyquist criterion is a theoretically sufficient condition to allow an analog signal to be reconstructed completely from a set of uniformly spaced discrete-time samples.

Ideal Sampling

- Ideal sampling can be represented with a sequence of unit impulse functions; also called Impulse Sampling.
- The sampling of $x(t)$ can be viewed as the product of $x(t)$ with a periodic train of unit impulse function $x_\delta(t)$ and defined as,

$$x_\delta(t) = \left[\sum_{n=-\infty}^{n=\infty} \delta(t - nT_s)\right] \qquad (1)$$

where T_s is the sampling period and $\delta(t)$ is the unit impulse or Dirac delta function.

- Resulting sampled data sequence,

$$x_s(t) = [x(t)x_\delta(t)]$$

Natural Sampling

- Multiply a band-limited analog signal $x(t)$ with a pulse train $x_p(t)$ or switching waveform.
- Resulting sampled data sequence,

$$x_s(t) = \left[x(t)x_p(t)\right] \tag{2}$$

- It is called so, since the top of each pulse in the $x_s(t)$ sequence retains the shape of its corresponding analog segment during the pulse interval.

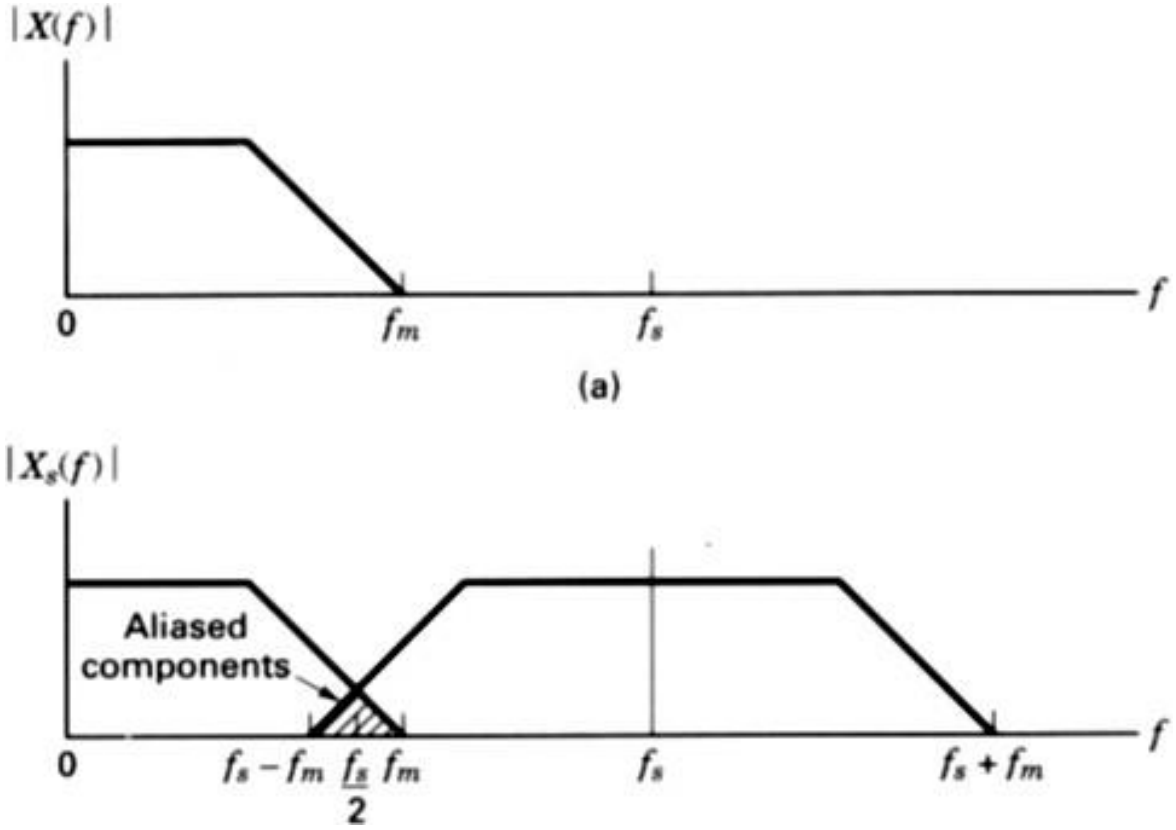
Flat-top Sampling

The simplest and most popular sampling method, sample and hold, can be described by the convolution of the sampled signal, $x(t)\mathrm{x}_{\delta}\ (t)$, with a unity amplitude rectangular pulse $p(t)$ of pulse width T_s. This time, convolution results in flat-top sampled sequence,

$$x_s(t) = p(t) * [x(t)x_{\delta}(t)] = p(t) * \left[x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)\right] \qquad (3)$$

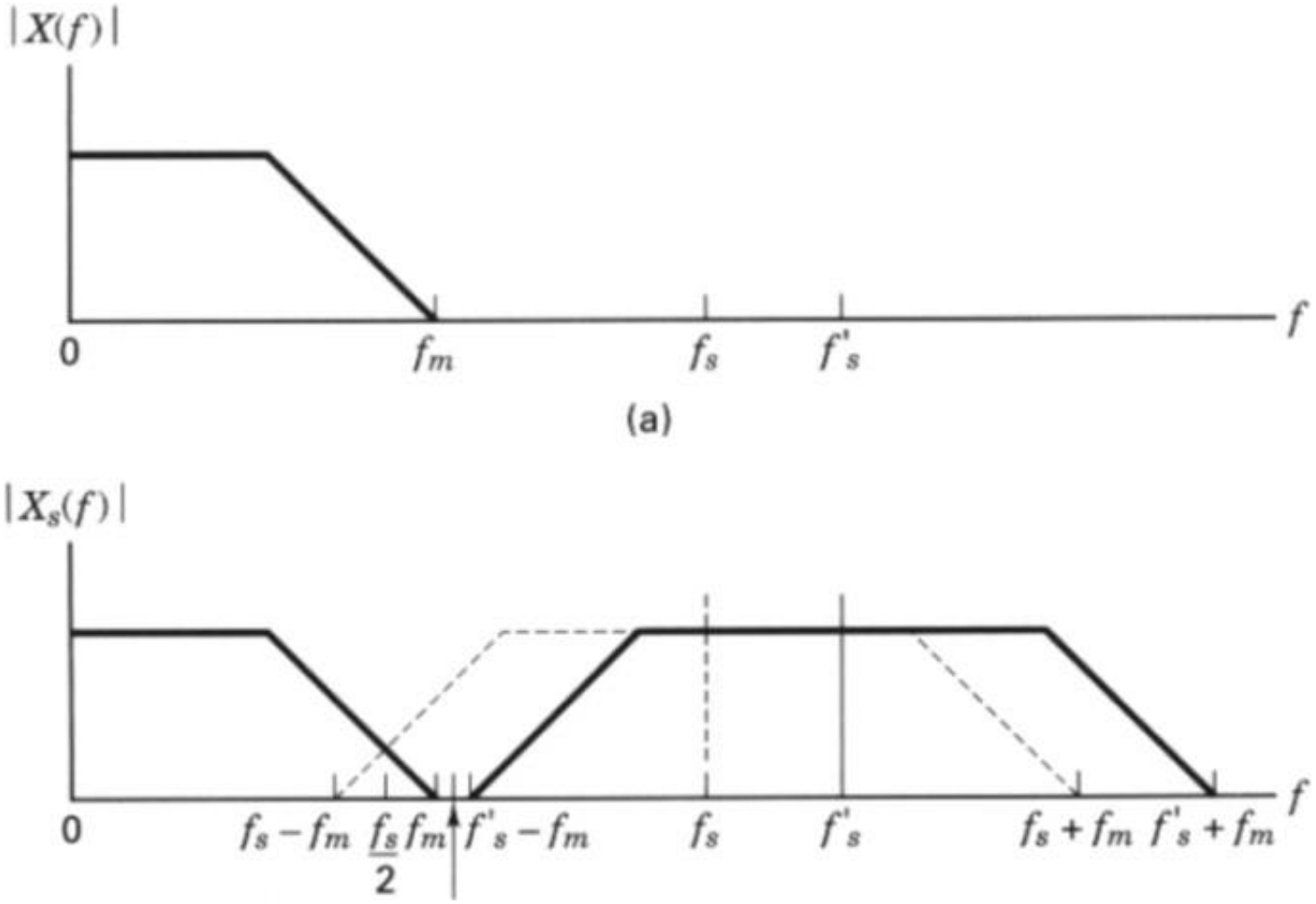
Aliasing

- Effect that causes different signals to become indistinguishable resulting from the distortions, when a signal is reconstructed from samples, and it is different from the original continuous signal.
- Under-sampling causes aliasing.



Aliasing

Over-sampling can eliminate aliasing.



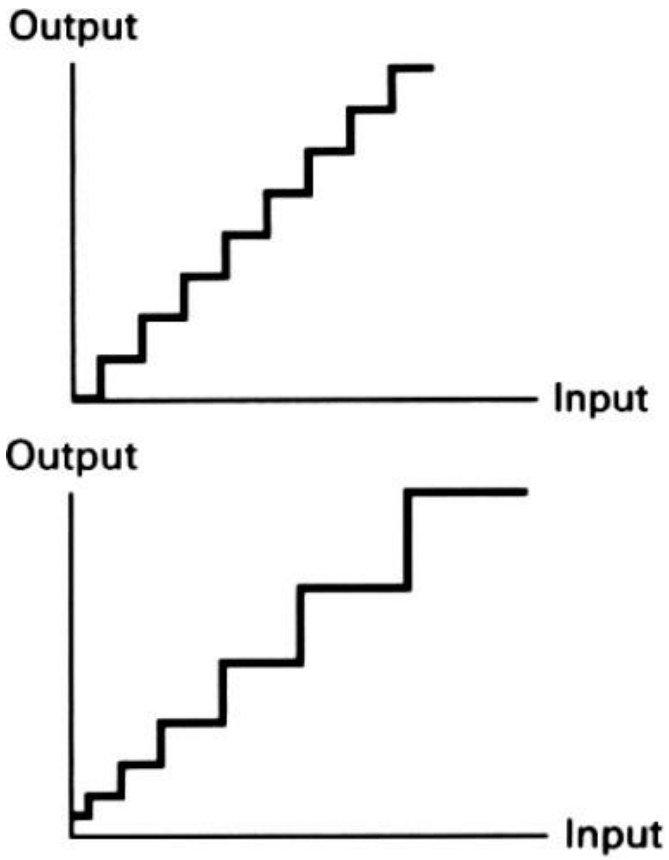
Quantization

- Sampling results in a series of pulses of varying amplitude values ranging between two limits: a min and a max.
- We need to map the infinite amplitude values onto a finite set of known values. This is achieved by dividing the distance between min and max into L zones, each of height Δ, $\Delta = \frac{\max - \min}{L}$
- The midpoint of each zone is assigned a value from *0* to *L−1*. Each sample falling in a zone is then approximated to the value of the midpoint.
- Each zone is then assigned a binary code. The number of bits required to encode the zones, or the number of bits per sample is obtained as follows: $n_b = \log_2 L$

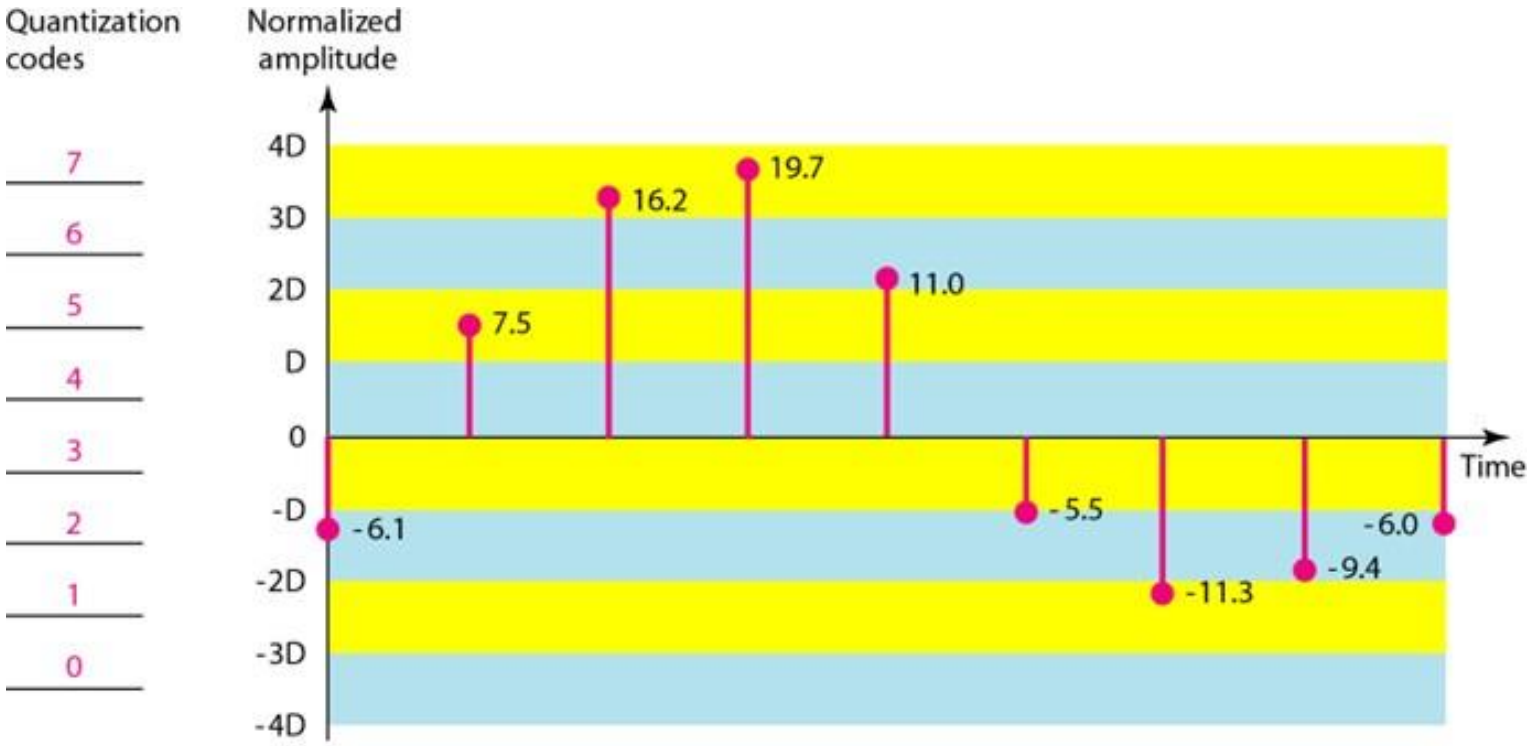
Types of Quantization

- *Uniform Quantization* – Steps are of equal size, equally spaced quantizer levels; wasteful for speech signals, many of the quantization levels will never be used.

- *Non-uniform Quantization* – Steps are of different sizes; fine quantization of weak signals, coarse quantization of strong signals.



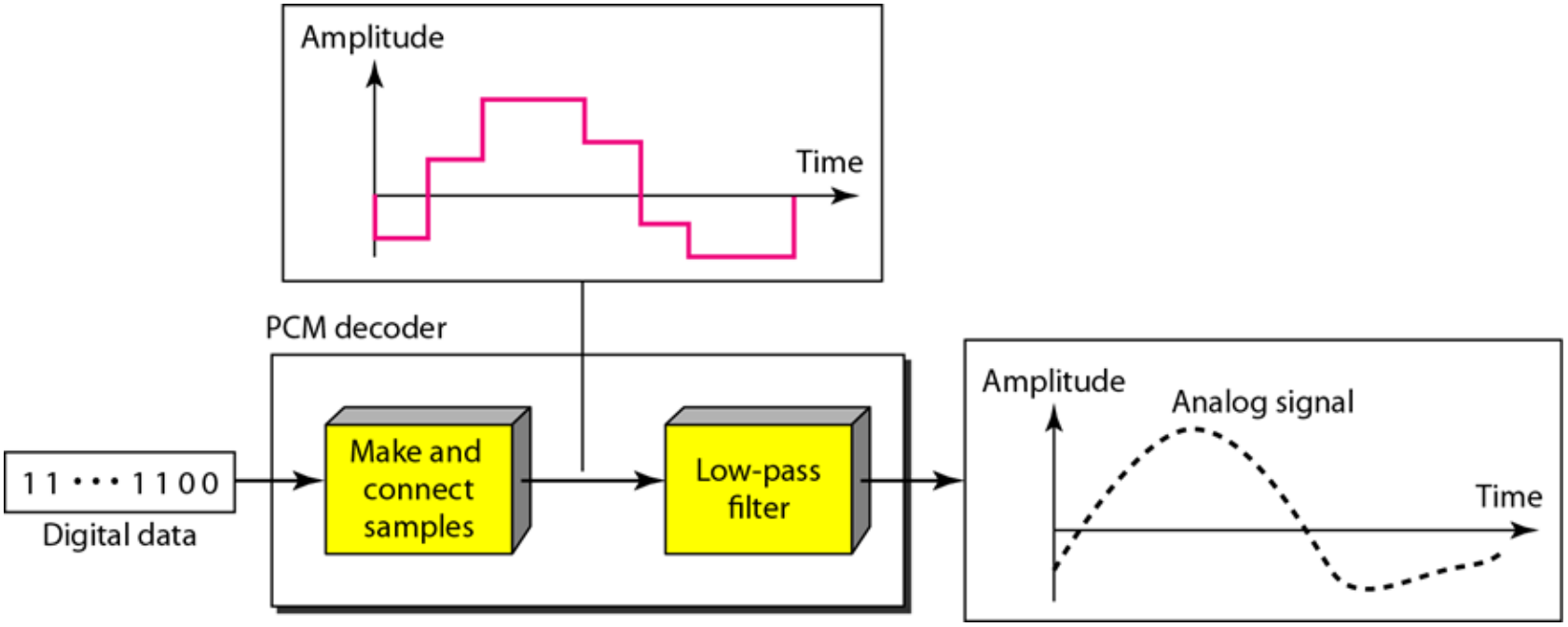
Quantization and Encoding



Normalized PAM values	-1.22	1.50	3.24	3.94	2.20	-1.10	-2.26	-1.88	-1.20
Normalized quantized values	-1.50	1.50	3.50	3.50	2.50	-1.50	-2.50	-1.50	-1.50
Normalized error	-0.38	0	+0.26	-0.44	+0.30	-0.40	-0.24	+0.38	-0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

PCM Decoder

- We use a hold circuit that holds the amplitude value of a pulse till the next pulse arrives.
- We pass this signal through a low pass filter with a cutoff frequency that is equal to the highest frequency in the pre-sampled signal.



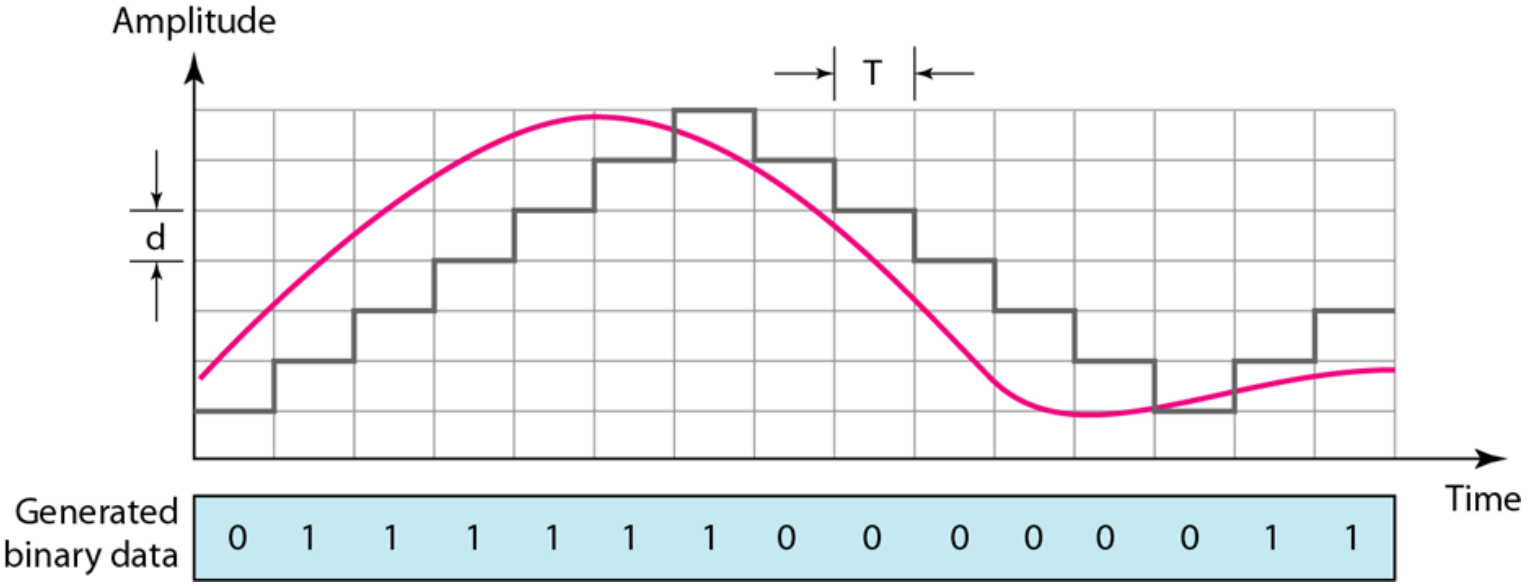
Sources of Corruption

Sources of corruption in the recovered analog signal,

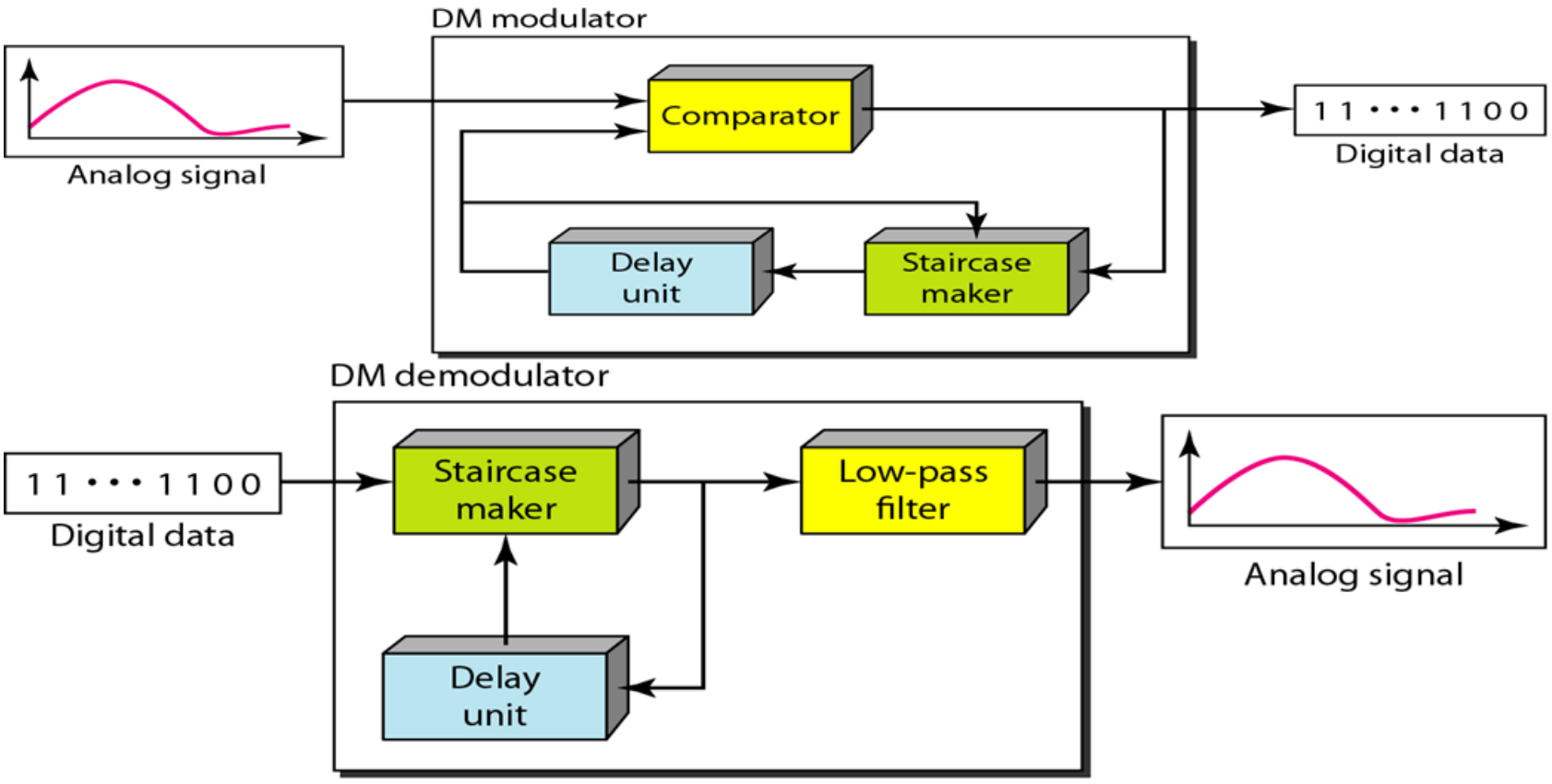
- ***Quantization Noise*** – Distortion introduced by the need to approximate the analog waveform with quantized samples.
- ***Quantization Saturation*** – The quantizer allocates L levels to the task of approximating the continuous range of inputs with a finite set of outputs. The range of inputs for which the difference between input and output is small is called the operating range of the converter. If the input exceeds this range, the converter is operating in saturation.
- ***Timing Jitter*** – If there is a slight jitter in the position of the sample, the sampling is no longer uniform.

Delta Modulation

This scheme sends only the difference between pulses, if the pulse at time t_{n+1} is higher in amplitude value than the pulse at time t_n, then a single bit, say a "1", is used to indicate the positive value. If the pulse is lower in value, resulting in a negative value, a "0" is used.



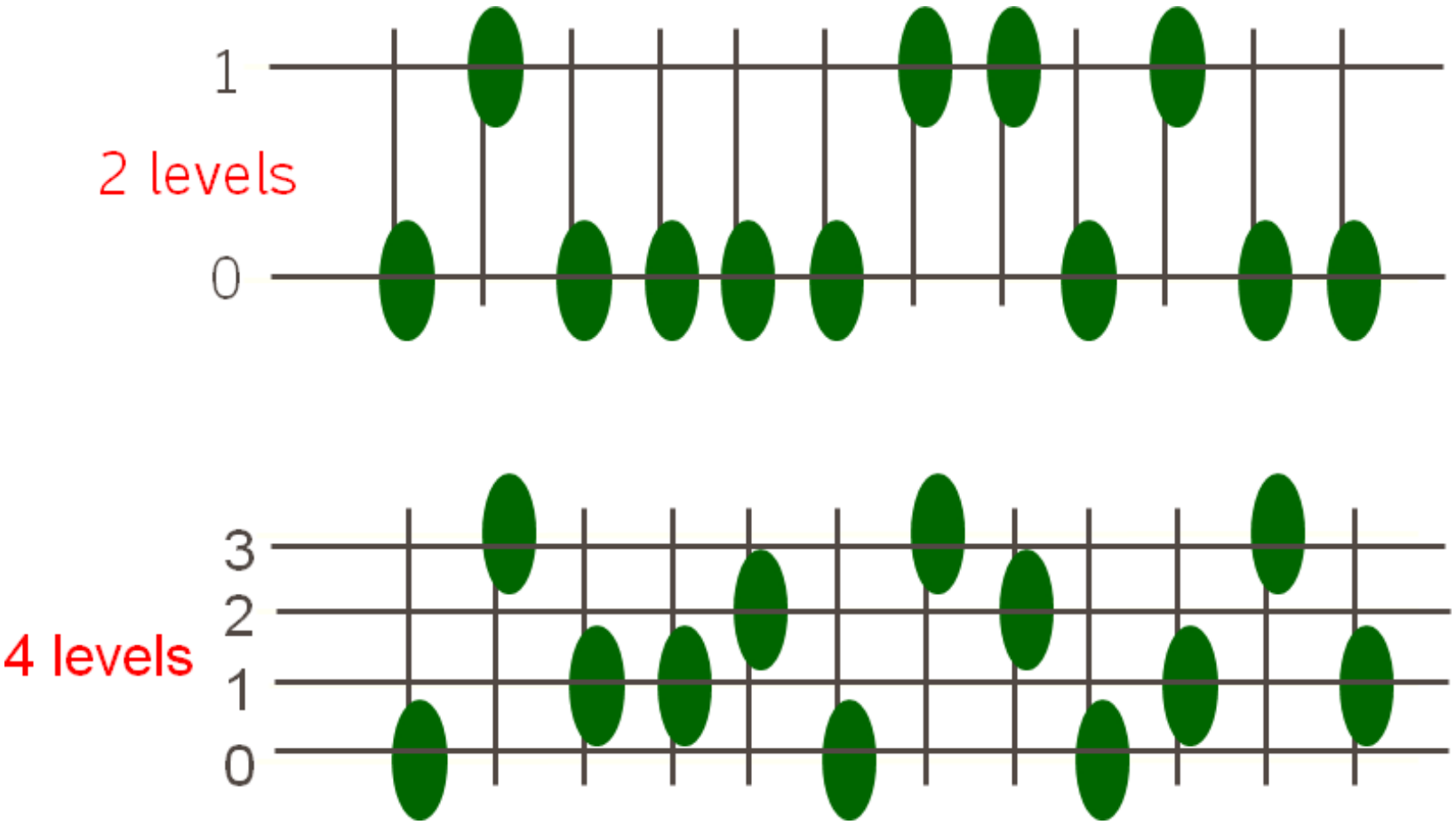
DM Modulator and Demodulator



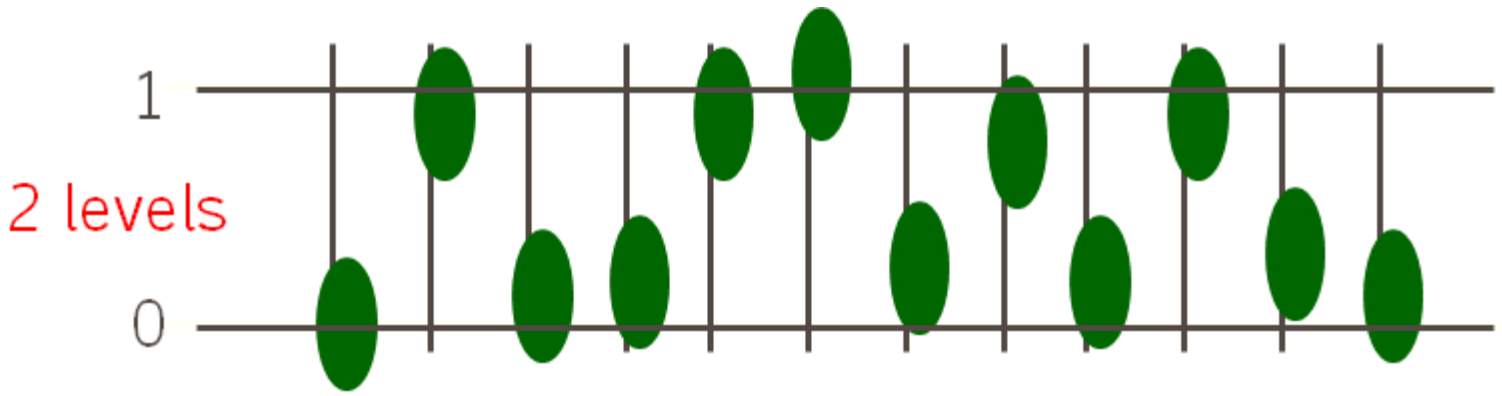
Problems with PCM and DM

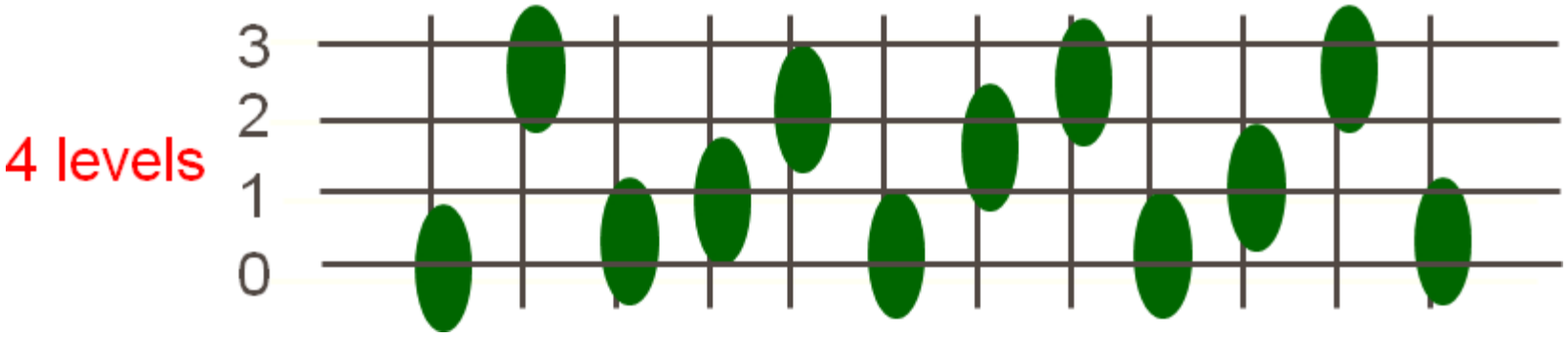
- The transmission bandwidth requirement for binary digital waveforms like PCM/DM is large.
- One possibility is **Multilevel Signalling**.
- In this case, data is partitioned into k bits groups.
- $M = 2^k$ level pulses are used for transmission.

What's wrong with multi-levels??



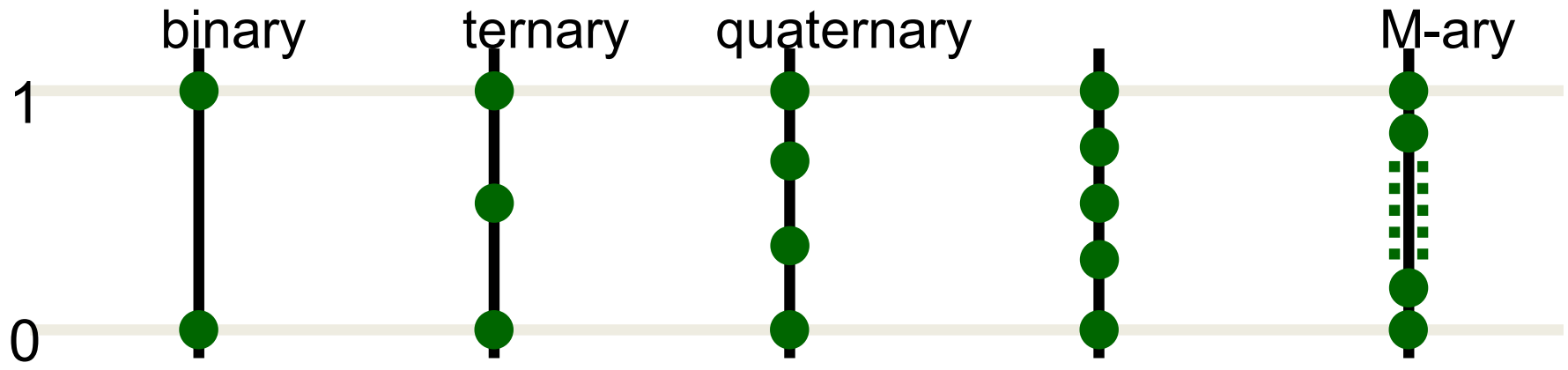
Give me the value of thesesignals





Baseband Digital Modulation

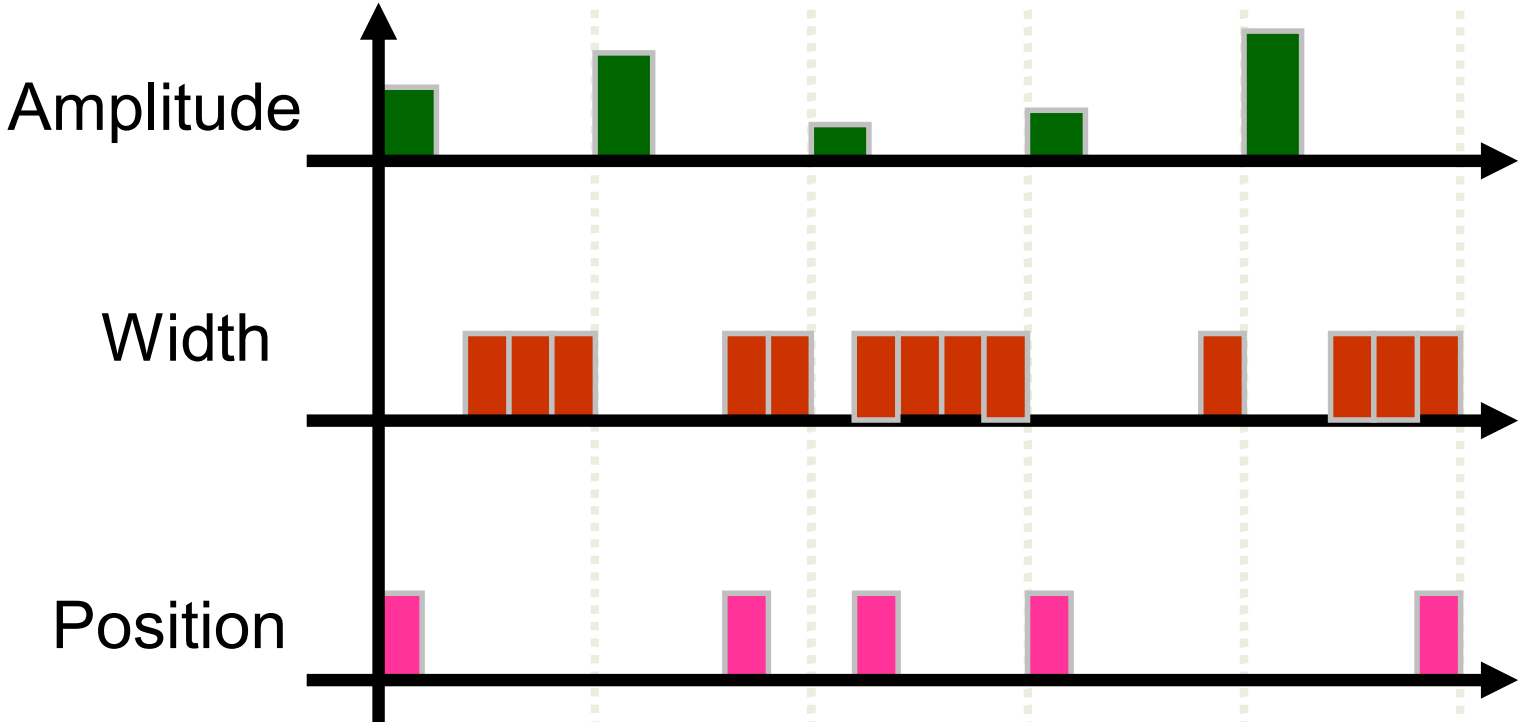
- The baseband modulator takes the binary signal at its input and embeds it in a discrete M -ary signal, where M is the finite number of values that the signal can take also called "symbols"
- Binary (2 values) → 1 bit per symbol
- Quaternary (4 values) → 2 bits per symbol



- Obviously, we can transfer more information if we use more steps (symbols), so why not use them?

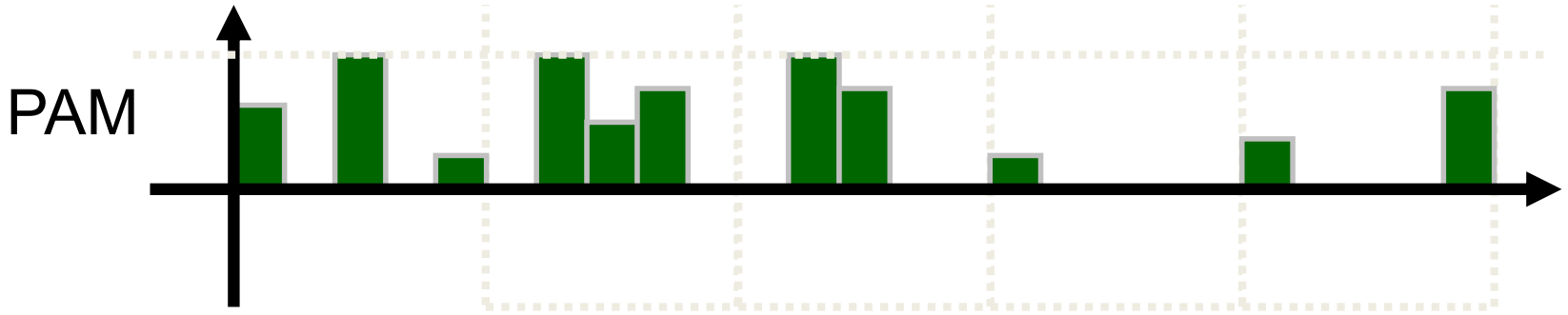
Pulse Digital Modulation

The output signal from the baseband digital modulator is formed by square pulses. Depending on the type of modulation, one different property of the pulses is varied to embed the symbol value. Typically we vary the amplitude, width, position, or density of pulses in a period of time



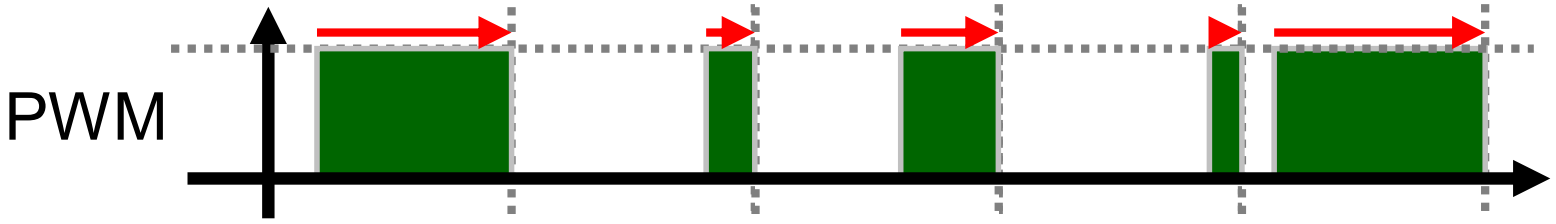
Pulse Amplitude Modulation (PAM)

- In a normally encoded system, as we normally understand it, the data consists of a sequence of pulses of pre-agreed height and of pre-agreed location (timing) and width.
- So each pulse is of fixed width and comes at an expected agreed time, but the amplitude varies. A special case is binary where the signal is either a fixed height or zero.
- The amplitude of the pulse corresponds to a valid level in our agreed *M*-ary system, in a binary system there are only 2 levels In a multi-level system, it can be at other amplitudes.



Pulse Width Modulation (PWM)

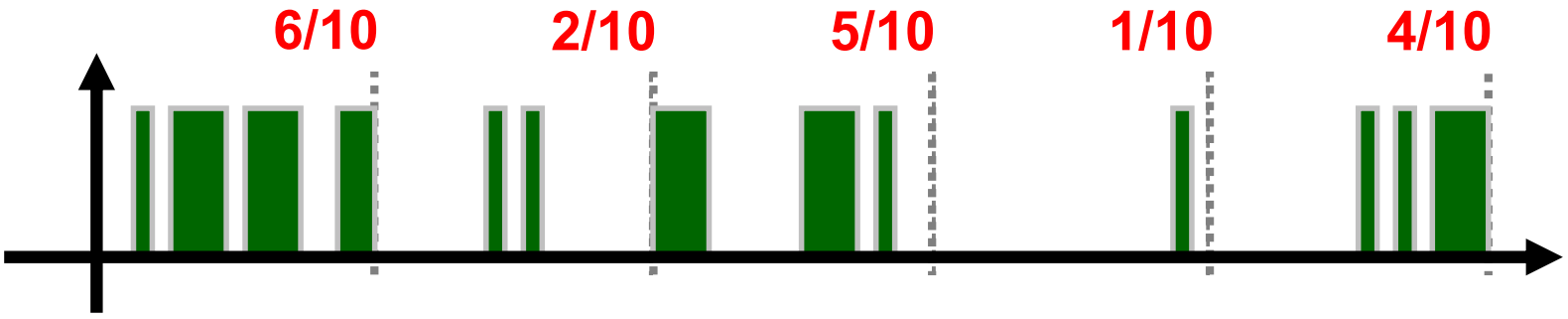
- The value of the signal is determined by the width of the pulse within a given time frame. There will only be one pulse during that time frame, and it will either start or finish with that pulse.
- The pulse is of fixed amplitude and so again suits systems with poor modulation control (on/off). The width of the pulse could be varied as an analogue signal, but the width variations are normally discrete values.
- Very simple scheme to implement and detect. To get the value, just calculate the time up and divide by the total time. The agreement to start/finish with a pulse makes it very easy to detect the pulse.



Pulse Density Modulation (PDM)

- The value of the signal is determined by the density of pulses within a given time frame. The pulses may be of varying time and width, but normally are of fixed amplitude. The value is extracted by determining the rolling average of the pulses.
- PWM is a special case of PDM where we have agreed to only have one pulse per time period. The scheme is more difficult to implement. What you need to do is calculate a rolling average and that average is the signal. The length of your rolling average and how often you look at the value of your average will determine how many levels you can represent.

PDM



Baseband Digital Demodulation

- On the receiver side, the detector is in charge of demodulating the signal, this means translate the received pulses or symbols into binary code.
- As it was seen in the previous slide, the higher the noise level, the higher the probability that the detector will make a mistake.
- In general, digital communication systems use other coding techniques (channel coding) which are used on the receiver side to reduce the number of mistakes made by the detector. You will see this next year.
- Finally, a digital to analogue converter provides the information signal to the sink in its original analogue format.